

Data Inspection Report

1. Dataset Overview

- Source: Telco Customer Churn Dataset (Kaggle)
- Total records: 7,043
- Total columns: 21 original (26 after your calculated columns)
- Granularity: one row per customer

2. Column Inventory

Briefly list the 21 original columns grouped by type:

- **Identifiers:** customerID
- **Demographics:** gender, SeniorCitizen, Partner, Dependents
- **Account info:** tenure, Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges
- **Services:** PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies
- **Target variable:** Churn

3. Data Quality Findings

- **Missing values:** 11 rows (0.16%) had blank TotalCharges values
- **Root cause:** all 11 rows had tenure = 0 — brand-new customers not yet billed
- **Duplicates:** 0 duplicate rows found
- **Data type issues:** TotalCharges was stored as text due to the blank entries; SeniorCitizen was stored as 0/1 instead of Yes/No

4. Cleaning Actions Taken

- Converted TotalCharges to numeric (Decimal Number) and filled the 11 blanks with 0, since tenure=0 means no charges have accrued yet — a logical zero, not an unknown value
- Converted SeniorCitizen from 0/1 to Yes/No for consistency with other binary fields
- Verified all remaining column types matched their expected format (text categories, whole numbers, decimals)

5. Summary Statistics (worth a small table)

- Average tenure: 32.4 months
- Average MonthlyCharges: \$64.76
- Churn rate: 26.5%